

Square Launch Hardening — Sprint Deliverable

Date: 2026-07-15

Goal: Prevent Open Order from collecting payment for Square-routed carts unless every ordered entity is routable at the vendor’s selected Square location. Eliminate the Poke Sea class of paid-but-not-routed failures.

Verdict

Launch-critical prevention for **partial mapping coverage** and **cross-location drift** is implemented and covered by tests. Square vendors with incomplete mappings at the selected location are **not orderable**, and checkout **blocks before payment** with `SQUARE_CART_PREFLIGHT_FAILED` .

Files changed

File	Purpose
<code>src/lib/integrations/square/square-mapping-coverage.server.ts</code>	Full sellable-menu coverage routability
<code>src/lib/integrations/square/square-order-routing-readiness.ts</code>	Replaced “any mapping” with readiness
<code>src/lib/integrations/square/square-cart-preflight.server.ts</code>	Pre-payment cart preflight for
<code>src/lib/integrations/square/square-routing-failure.ts</code>	Normalized failure payload + message
<code>src/lib/integrations/provider-mapping.service.ts</code>	<code>deactivateSquareMappings</code>
<code>src/lib/integrations/square/square-connection.service.ts</code>	Location change quarantine single-active upsert
<code>src/lib/integrations/square/square-menu-import.service.ts</code>	Clears <code>menuRequiresRepubl</code> successful import sync
<code>src/lib/vendor-order-routing-mode.ts</code>	<code>isVendorSquareOrderable</code>
<code>src/lib/vendor-readiness-states.ts</code>	<code>square_routing</code> operations
<code>src/lib/vendor-readiness-validation.server.ts</code>	Loads <code>squareOrderRoutingF</code> injection readiness
<code>src/lib/vendor-pod-readiness.ts</code>	Merges Square readiness flags <code>posSummary</code>
<code>src/services/order.service.ts</code>	Runs Square cart preflight in <code>validateCartForOrder</code>
<code>src/services/square-order.service.ts</code>	Persists structured <code>routingF</code> readiness/mapping fail

src/actions/vendor-square-connect.actions.ts	Actor + routing readiness for
src/actions/admin-square-mapping.actions.ts	Admin deactivate stale other mappings
src/components/vendor/VendorSquareConnectionCard.tsx	Coverage + non-orderable r location-change warning
src/app/vendor/.../integrations/square/page.tsx	Passes routing readiness into
src/app/admin/.../AdminSquareOrderInjectionDiagnosticsPanel.tsx	Coverage + stale-mapping d
src/app/admin/.../AdminDeactivateStaleSquareMappingsButton.tsx	Admin UX for quarantining o mappings
src/lib/integrations/square/admin-square-order-injection-diagnostics.server.ts	Exposes mappingCoverage
src/lib/admin-audit-log.ts	SQUARE_LOCATION_CHANGED, SQUARE_STALE_MAPPINGS_DE
prisma/migrations/20260715140000_square_one_active_connection/migration.sql	Partial unique index: one acti connection per vendor
prisma/schema.prisma	Documents partial unique inc
Tests under src/lib/integrations/square/*, vendor readiness/orderability, order cart validation, square-order.service	Launch-critical coverage

Schema and migrations

Added:

prisma/migrations/20260715140000_square_one_active_connection/migration.sql

```
CREATE UNIQUE INDEX IF NOT EXISTS
"VendorIntegrationConnection_one_active_square_per_vendor"
ON "VendorIntegrationConnection" ("vendorId", "provider")
WHERE "isActive" = true AND "provider" = 'square';
```

No new columns. Historical inactive Square connection rows remain allowed.

Apply with your usual migrate deploy path before production enablement.

Readiness behavior before and after

	Before	After
Item mappings	activeItemMappingCount > 0	Every sellable operational available item mapped at selected location

gate		
Cross-location maps	Counted only at selected location (stale old-location maps ignored for count but left active)	MAPPING_AT_DIFFERENT_LOCATION blockers; not ready
Required modifiers	Not required for readiness	Required groups/options must be mapped at selected location
Optional modifiers	N/A	Not required until selected (cart preflight checks selections)
Structured result	Counts + string blockers	mappingCoverage + coverageBlockers

prerequisitesReady = connection + scopes + location + Square-published menu + **coverage ready**.
 injectionOperationalReady = prerequisites + SQUARE_ROUTING_LIVE .
 Public orderability uses squareOrderRoutingReady = injectionOperationalReady .

Checkout behavior before and after

	Before	After
validateCartForOrder	Orderability + Deliverect kitchen mapping	Same + validateSquareCartPreflight
Payment Intent	After order create from cart	Still after order create — validation already failed if Square cart unroutable
Failure code	N/A	SQUARE_CART_PREFLIGHT_FAILED
Customer message	N/A	"One or more items are no longer available from this vendor..."
Multi-vendor	Sibling Square vendor could fail post-pay	Any Square vendor failing preflight blocks the whole checkout before charge

Preflight does **not** call Square CreateOrder/CreatePayment.

Location-change behavior

On selectSquareLocationForVendor when location **changes**:

1. Transaction updates connection location + sets menuRequiresRepublish: true .
2. Deactivates all active Square mappings for that vendor **outside** the new location.
3. Audit log SQUARE_LOCATION_CHANGED with previous/new location and deactivated count.
4. Vendor immediately fails coverage readiness until re-import + publish at the new location.

Successful catalog import sync clears menuRequiresRepublish .

Test results

Command:

```
npx vitest run \  
  src/lib/integrations/square/square-mapping-coverage.test.ts \  
  src/lib/integrations/square/square-order-routing-readiness.test.ts \  
  src/lib/integrations/square/square-cart-preflight.test.ts \  
  src/lib/integrations/square/square-location-mapping-quarantine.test.ts \  
  src/lib/integrations/square/square-connection.service.test.ts \  
  src/lib/integrations/square/admin-square-order-injection-diagnostics.test.ts \  
  src/lib/vendor-readiness-states.test.ts \  
  src/lib/vendor-order-routing-mode.test.ts \  
  src/lib/vendor-readiness-validation.server.test.ts \  
  src/lib/vendor-pod-readiness.test.ts \  
  src/services/square-order.service.test.ts \  
  src/services/order.service.cart-validation.test.ts
```

Result: 12 files, 138 passed.

Tests added (high signal)

- Coverage: zero / partial / full / unavailable ignored / Poke Sea wrong-location / required mod / optional mod / inactive / other vendor scope / location unset
- Cart preflight: readiness fail, multi-vendor one-fail, modifier fail, non-Square pass-through
- Location quarantine deactivates outside selected location
- Orderability: visible but closed when `squareOrderRoutingReady` false
- Bundles load `squareOrderRoutingReady` ; incomplete coverage blocks orderability

Remaining risks (intentionally out of sprint)

- Square refund / cancel money synchronization
- Background status polling when webhooks are silent
- OO → Square catalog export
- Full Kitchen Mode nav cleanup
- Long-term Square sales reporting reconciliation
- Concurrent OAuth callback race under extreme load (mitigated by partial unique index + transactional deactivate)

Acceptance criteria checklist

- ☒ Square vendor not orderable with partial menu mapping coverage
- ☒ Mapping at a different Square location never satisfies readiness
- ☒ Changing Square location immediately makes vendor non-orderable (coverage empty + quarantined maps)

- ☒ Old-location mappings cannot be used for new orders
- ☒ Exact cart validated against selected location before payment (`validateCartForOrder`)
- ☒ Unroutable Square cart cannot be charged (preflight failure)
- ☒ Multi-vendor checkout cannot charge when any Square vendor fails preflight
- ☒ Reconnect path deactivates other actives; DB unique partial index enforces one active
- ☒ Structured diagnostics persisted on readiness/mapping submit failures (`routingFailure`)
- ☒ Poke Sea scenario reproduced in tests and prevented
- ☒ Manual/Deliverect paths unchanged in targeted tests
- ☒ Relevant tests pass (138)